

AMS Bootcamp Latex Workshop

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Wednesday
August 20th, 2025

Outline

- 1 Introduction
- 2 Homeworks
- 3 Posters
- 4 Presentations
- 5 Manuscripts

What is it?

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Typesetting software originally written in the 1980s.

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Why people love it:

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- Minimum and flexible citation management.
- Typing math.
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Why people hesitate to use it:

- ◇ Steep learning curve when getting started, good templates?
- ◇ Overleaf renders too slowly and sometimes full of bugs.
- ◇ Different bibliography options and compilers are confusing.
- ◇ Working with scientists that prefer Word or Google Doc.

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- Great for thesis, projects, books, etc.

Agenda

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① Setup

Go to GitHub link: github.com/ziyuli22/AMS_Latex_Workshop and download folder. Log into Overleaf account, upload folder.

OR

Download this folder from Canvas. Log into Overleaf, upload folder.

② Common homework commands & expectations

③ Poster example

④ Presentation example

⑤ Manuscripts Tips

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Template Example

Go to Homework_Example folder.
Click on `Homework_Example.tex`

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Poster Example

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Click on **Poster_Example.tex**

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Take a look at

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This presentation is a modification on the Madrid theme.

Add Algorithms & Use Pause

Data: $\mathbf{y}$ at n spatial locations $\mathcal{S}^O = \{\mathbf{s}_1^O, \mathbf{s}_2^O, \dots, \mathbf{s}_n^O\}$.

Goal: Quantify uncertainty of predictions $\hat{g}$ on evenly spaced grid $\mathcal{S}^G = \{\mathbf{s}_1^G, \mathbf{s}_2^G, \dots, \mathbf{s}_M^G\}$.

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Algorithm 1 Conditional Simulation Method

Input: Spatial data $\mathbf{y}$, their locations $\mathcal{S}^O$, and prediction grid locations $\mathcal{S}^G$.

Output: Ensemble of l conditional simulations $\mathbf{v} = \{\mathbf{v}_1, \dots, \mathbf{v}_l\}$. $\mathbf{v}_j \sim MVN(\hat{g}(\mathcal{S}^G), \Sigma_{\hat{g}})$.

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For $j = 1 : l$

- Simulate spatial process at the union of locations $\mathcal{S}^S = \mathcal{S}^G \cup \mathcal{S}^O$, label this $g^S(\mathcal{S}^S)$.

End

Others

- Everything else is similar to poster. Be mindful how large your picture files are because that can slow down rendering.
- Make sure to cite things too [1].

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Manuscripts Overview

- Most journals have templates that you can use.
 - EVERYTHING as files in one folder, including figures.
 - Be mindful of file management and clear naming.
 - Special commands from journal template, and sources management can be tricky.
- Slightly easier to debug and upload to journal or ArXiv if you use off-line compiler (imo).
- Be mindful of readability and use safe colors.

References

- [1] Aurthor01 CoolLastName, Author02 AnotherLastName, and Aurthor03 ALastName. “Place Holder Title for a Fake Journal”. In: *Place Holder Journal* 11 (1 Dec. 2022). issn: 2000000. doi: 000000000.

Extra Slides for Questions or other Technical Details

Simulation via Cholesky Decomposition

Cholesky decomposition: $\Sigma = BB^T$

Obtain simulation: $g(\mathbf{s}) = B\boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \text{MVN}(0, I)$

$$\mathbb{E}[B\boldsymbol{\epsilon}] = B\mathbb{E}[\boldsymbol{\epsilon}] = 0$$

$$\text{Var}(B\boldsymbol{\epsilon}) = B\text{Var}(\boldsymbol{\epsilon})B^T = BIB^T = \Sigma$$

Sometimes you include more slides than you actually present to be prepare to demonstrate difficult concepts.